The terms proportional to $\log \frac{k^+}{\Lambda}$ (JIMWLK) and $\log \frac{\vee}{k^+}$ (BFKL), row by row from `Group_I/II/III.tex` and `4rho.tex`

Notation as in your files: $\mathcal{K}(X) = X/X^2$, $K(a,b,z) = \mathcal{K}(a-z)\cdot\mathcal{K}(b-z)$, $U^{bc} = U^{cb}$,

$$\mathcal{N} = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}.$$

Blue: new or corrected; red: removed. The finite terms (everything that is not $\propto \log\frac{k^+}{\Lambda}$ or $\log\frac{\vee}{k^+}$) are in the separate file `Finite_terms_rows.pdf`.

1 Answer

Every row is split exactly, term by term,

$$\frac{d^3N_{\rm row}}{d^3k} = \log\frac{k^+}{\Lambda}\,\mathfrak{T}_{\rm row} + \log\frac{\vee}{k^+}\,\mathfrak{P}_{\rm row} + R_{\rm row}, \tag{1.1}$$

with $\log\frac{\vee}{\Lambda} = \log\frac{k^+}{\Lambda} + \log\frac{\vee}{k^+}$ and the splits of Sec. 2; $\vee > p^+$, $k^+ > \Lambda$; $\Lambda \to 0$ in $R_{\rm row}$ wherever possible, $\vee$ exact. Numerically, for every one of the 79 terms that can be evaluated pointwise, your term = its three parts to 1.5×10^{-14}, and the remainder has its $\Lambda\to0$ limit (`check_split.py`).

- **Three** ρ**'s** $\propto \log\frac{k^+}{\Lambda}$ (22 terms): **cancel**, relative residual 7.7×10^{-16} (Sec. 7).
- **Three** ρ**'s** $\propto \log\frac{\vee}{k^+}$ (35 terms): **cancel**, 1.5×10^{-16}.
- **Four** ρ**'s** (your `4rho.tex`): they have $\log\frac{\vee}{\Lambda}$, i.e. the same coefficient for both logarithms. They cancel after symmetrization (Sec. 6); their three- and two-ρ remainders enter both lists.
- **Two** ρ**'s** $\propto \log\frac{k^+}{\Lambda}$ = **JIMWLK⊗LO**, exactly: 13 kernel structures, 154 independent colour structures, 0 differences (Sec. 8).
- **Two** ρ**'s** $\propto \log\frac{\vee}{k^+}$ = **BFKL⊗LO**, exactly: 9 kernel structures, 109 colour structures, 0 differences (Sec. 9).

Therefore

$$\frac{d^3N_{\rm LL}}{d^3k} = \mathcal{N}\Big[\log\frac{k^+}{\Lambda}\ \text{JIMWLK}\otimes\text{LO} + \log\frac{\vee}{k^+}\ \text{BFKL}\otimes\text{LO}\Big], \tag{1.2}$$

with the targets of Sec. 10. The target evolves (JIMWLK) between $y_T = \log\Lambda$ and the gluon, the projectile (BFKL) between the gluon and $y_P = \log\vee$.

What has to change in your files (shown in red/blue in the rows):

E1 Group I Row V 2b UV: the bracket must have $2\log\frac{k^+}{\Lambda}$ (from $2\log\frac{\vee}{\Lambda} - 2\log\frac{\vee}{k^+}$; the factor 2 is lost after "Setting $\Lambda\to0$"). With your coefficient 1 the poles $\times\log\frac{k^+}{\Lambda}$ do not cancel (Sec. 3, Row V).

F3 Group I Row III d^3N_{3a}: the logarithmic line with the edge kernel is replaced by $\mathcal{D}$. Without it the three-ρ terms $\propto\log\frac{k^+}{\Lambda}$ do not cancel (5.9×10^{-1}).

H Hidden logarithms. In Group III Rows III (both parts), V, VI and Group II Row IV the $\log\frac{\vee}{k^+}$ inside $\big[\log\frac{k^+}{\Lambda}+\log(\frac{\vee}{k^+}-1)\big]$ is *not* a BFKL term: it cancels against the one inside $\big[\frac{1}{p^+}\big]_+$ (resp. $\frac{\xi}{(1-\xi)_+}$) of the same row. The BFKL terms of these rows are the terms (B) of `BFKL_final.pdf` and H_4 (blue). Their $\log\frac{k^+}{\Lambda}$ parts are as you have them.

C5 Group III Row III part 2: the measure is $\int_{w,x',y',z,x}$.

F1, F2 The two-ρ lists of `4rho.tex` still need the two fixes of `Final_evolution_check.pdf` (Sec. 8).

M2 Your JIMWLK formula (App. F) needs the correction of Sec. 10; your BFKL formula is correct.

Everything else is as you wrote it.

2 How the logarithms are split

A term carries at most one logarithmic factor. Your brackets:

$$\left[\log\left(\frac{k^+}{\Lambda}\right)+\log\left(\frac{\vee}{k^+}-1\right)\right]=\log\frac{k^+}{\Lambda}+\log\frac{\vee}{k^+}+\log\left(1-\frac{k^+}{\vee}\right),$$
$$\left[\log\left[\frac{\vee}{k^+}\right]+\log\left[\frac{k^+}{\Lambda}\right]\right]=\log\frac{k^+}{\Lambda}+\log\frac{\vee}{k^+},$$

except in the rows with a hidden logarithm (H above), where the first bracket contributes only $\log\frac{k^+}{\Lambda}$ and $\log(\frac{\vee}{k^+}-1)$ stays in the finite terms. All other logarithms ($A=(x-z)^2$ etc.; the remainder is given exactly and, where it depends on Λ, its limit):

$$\log\left(\frac{\vee}{k^+}-1\right)=\log\frac{\vee}{k^+}+\log\left(1-\frac{k^+}{\vee}\right)$$
$$\log\left(\frac{k^+-\Lambda}{\Lambda}\right)=\log\frac{k^+}{\Lambda}+\log\left(1-\frac{\Lambda}{k^+}\right)\xrightarrow{\Lambda\to0}\log\frac{k^+}{\Lambda}$$
$$\log\left[\frac{k^+-\Lambda}{\Lambda}\right]=\log\frac{k^+}{\Lambda}+\log\left(1-\frac{\Lambda}{k^+}\right)\xrightarrow{\Lambda\to0}\log\frac{k^+}{\Lambda}$$
$$\log\left[\frac{\Lambda}{k^+-\Lambda}\right]=-\log\frac{k^+}{\Lambda}-\log\left(1-\frac{\Lambda}{k^+}\right)\xrightarrow{\Lambda\to0}-\log\frac{k^+}{\Lambda}$$
$$\log\left[\frac{\vee}{\Lambda+k^+}\right]=\log\frac{\vee}{k^+}-\log\left(1+\frac{\Lambda}{k^+}\right)\xrightarrow{\Lambda\to0}\log\frac{\vee}{k^+}$$
$$\log\left[\frac{\vee}{k^++\Lambda}\right]=\log\frac{\vee}{k^+}-\log\left(1+\frac{\Lambda}{k^+}\right)\xrightarrow{\Lambda\to0}\log\frac{\vee}{k^+}$$
$$\log\left[\frac{\left[\vee(x-z)^2-k^+(x-w)^2\right]}{\left[(\Lambda+k^+)(x-z)^2-k^+(x-w)^2\right]}\right]=\log\frac{\vee}{k^+}+\log\left[\frac{(x-z)^2-\frac{k^+}{\vee}(x-w)^2}{\left(1+\frac{\Lambda}{k^+}\right)(x-z)^2-(x-w)^2}\right]\xrightarrow{\Lambda\to0}\log\frac{\vee}{k^+}+\log\left[\frac{(x-z)^2-\frac{k^+}{\vee}(x-w)^2}{(x-z)^2-(x-w)^2}\right]$$
$$\log\left[\frac{\left[(k^+-\Lambda)(x-z)^2-k^+(x-w)^2\right]}{\left[\Lambda(x-z)^2-k^+(x-w)^2\right]}\right]=\log\left[\frac{(k^+-\Lambda)(x-z)^2-k^+(x-w)^2}{\Lambda(x-z)^2-k^+(x-w)^2}\right]\xrightarrow{\Lambda\to0}\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]$$
$$\log\left[\frac{\left[k^+(x-w)^2-\vee(x-z)^2\right]}{\left[k^+(x-w)^2-\Lambda(x-z)^2\right]}\right]=\log\frac{\vee}{k^+}+\log\left[\frac{\frac{k^+}{\vee}(x-w)^2-(x-z)^2}{(x-w)^2-\frac{\Lambda}{k^+}(x-z)^2}\right]\xrightarrow{\Lambda\to0}\log\frac{\vee}{k^+}+\log\left[\frac{\frac{k^+}{\vee}(x-w)^2-(x-z)^2}{(x-w)^2}\right]$$
$$\log\left[\frac{(x-w)^2k^+-(x-z)^2\vee}{(x-w)^2k^+-(x-z)^2(k^++\Lambda)}\right]=\log\frac{\vee}{k^+}+\log\left[\frac{(x-w)^2\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\left(1+\frac{\Lambda}{k^+}\right)}\right]\xrightarrow{\Lambda\to0}\log\frac{\vee}{k^+}+\log\left[\frac{(x-w)^2\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]$$
$$\log\left[\frac{(x-w)^2k^+-(x-z)^2(k^+-\Lambda)}{(x-w)^2k^+-(x-z)^2\Lambda}\right]=\log\left[\frac{(x-w)^2k^+-(x-z)^2(k^+-\Lambda)}{(x-w)^2k^+-(x-z)^2\Lambda}\right]\xrightarrow{\Lambda\to0}\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]$$
$$\log\left[\frac{\left[k^+(x-w)-(k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w)-\Lambda(x-z)\right]^2}\right]=\log\left[\frac{\left[k^+(x-w)-(k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w)-\Lambda(x-z)\right]^2}\right]\xrightarrow{\Lambda\to0}\log\left[\frac{(z-w)^2}{(x-w)^2}\right]$$
$$\log\left[\frac{(\vee-k^+)(x-z)^2+k^+(x-w)^2}{\Lambda(x-z)^2+k^+(x-w)^2}\right]=\log\frac{\vee}{k^+}+\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{\frac{\Lambda}{k^+}(x-z)^2+(x-w)^2}\right]$$

$$\xrightarrow{\Lambda\to 0} \log\frac{\vee}{k^+} + \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\right]$$

$$\log\left[\frac{(\vee-k^+)(x-z)^2+k^+(x-w)^2}{k^+(x-w)^2+\Lambda(x-z)^2}\right] = \log\frac{\vee}{k^+} + \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{\frac{\Lambda}{k^+}(x-z)^2+(x-w)^2}\right]$$

$$\xrightarrow{\Lambda\to 0} \log\frac{\vee}{k^+} + \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\right]$$

$$\log\left[\frac{(\vee-k^+)(x-z)^2+k^+(x-w')^2}{\Lambda(x-z)^2+k^+(x-w')^2}\right] = \log\frac{\vee}{k^+} + \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w')^2}{\frac{\Lambda}{k^+}(x-z)^2+(x-w')^2}\right]$$

$$\xrightarrow{\Lambda\to 0} \log\frac{\vee}{k^+} + \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w')^2}{(x-w')^2}\right]$$

$$\log\left[\frac{(\vee-k^+)(x'-z)^2+k^+(x'-w')^2}{\Lambda(x'-z)^2+k^+(x'-w')^2}\right] = \log\frac{\vee}{k^+} + \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}(x'-w')^2}{\frac{\Lambda}{k^+}(x'-z)^2+(x'-w')^2}\right]$$

$$\xrightarrow{\Lambda\to 0} \log\frac{\vee}{k^+} + \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}(x'-w')^2}{(x'-w')^2}\right]$$

$$\log\left[\frac{k^+(x'-w')^2+(\vee-k^+)(x'-z)^2}{k^+(x'-w')^2+\Lambda(x'-z)^2}\right] = \log\frac{\vee}{k^+} + \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}(x'-w')^2}{\frac{\Lambda}{k^+}(x'-z)^2+(x'-w')^2}\right]$$

$$\xrightarrow{\Lambda\to 0} \log\frac{\vee}{k^+} + \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}(x'-w')^2}{(x'-w')^2}\right]$$

$$\log\left[\frac{(\vee-k^+)(x-z)^2+k^+(x-w)^2}{k^+(x'-w')^2+(\vee-k^+)(x'-z)^2}\right] = \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{\frac{k^+}{\vee}(x'-w')^2+\left(1-\frac{k^+}{\vee}\right)(x'-z)^2}\right]$$

$$\log\left[\frac{\Lambda(x-z)^2+k^+(x-w)^2}{k^+(x'-w')^2+\Lambda(x'-z)^2}\right] = \log\left[\frac{\Lambda(x-z)^2+k^+(x-w)^2}{k^+(x'-w')^2+\Lambda(x'-z)^2}\right]$$

$$\xrightarrow{\Lambda\to 0} \log\left[\frac{(x-w)^2}{(x'-w')^2}\right] \tag{2.1}$$

Logarithms of ϵ, μ, $\log(1-\frac{k^+}{\vee})$ and $\log(1-\frac{\Lambda}{k^+})$ are finite and are not split. Three special cases:

- Group I Row II part 1: $\left[\frac{11}{3}-2\log\frac{k^+}{\Lambda}+\log(1-\frac{k^+}{\vee})\right]$ times the pole bracket; only $-2\log\frac{k^+}{\Lambda}$ is large.
- Group I Row V 2b UV: $\left[2\log\frac{k^+}{\Lambda}+2\log(1-\frac{\Lambda}{k^+})-\frac{11}{6}\right]$ (E1).
- Group I Row V 2b finite, line 6: $\left[-\frac{11}{6}+2\log\frac{k^+-\Lambda}{\Lambda}\right]$.

Every line of every row below is in your form; only the logarithm is replaced by its large part.

3 Group I (with c.c.)

3.1 Row I

Only four ρ's: Sec. 6 (with the same coefficient for both logarithms).

3.2 Row II

Row II, first part. Your bracket $\left[\frac{11}{3}-2\log\frac{k^+}{\Lambda}+\log(1-\frac{k^+}{\vee})\right]$ multiplies the pole bracket; only its $-2\log\frac{k^+}{\Lambda}$ is a large logarithm. The term $\log(1-\frac{k^+}{\vee})\log\frac{\vee}{k^+}$ vanishes for $\vee\to\infty$ and is not a BFKL term; it stays with the finite terms. Terms $\propto\log\frac{k^+}{\Lambda}$:

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{32\pi^4}\,\frac{1}{k^+}\int_{w,w',x,x',} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[\left[-\frac{2}{\epsilon}-2\gamma-\log\left[\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\times\right.$$
$$\left.\left(-2\,\log\left(\frac{k^+}{\Lambda}\right)\right)\right]\times$$
$$\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ab}(w)\,\rho^{b}(x)-U^{ab}(x)\,\rho^{b}(x)\right] \tag{3.1}$$

Terms $\propto\log\frac{\vee}{k^+}$: none.

Row II, second part, d^3N_{2a}. Line 1 has $\left[\log\frac{\vee}{k^+}+\log\frac{k^+}{\Lambda}\right]$ and enters both lists. In lines 2 and 3 the z integral is already done (hence $\frac{1}{32\pi^4}$); the measure should read $\int_{x,y,x'}$. Terms $\propto \log\frac{k^+}{\Lambda}$:

$$\begin{aligned}
\frac{d^3N_{2\,a}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{hbc}}{32\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\,\frac{(w-z)\cdot(y-z)}{(w-z)^2\,(y-z)^2}\right.\\
&\left.-\,\frac{(y-w)^i}{(y-w)^2}\,\frac{(w-z)\cdot(x-z)}{(w-z)^2\,(x-z)^2}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\right]\\
&-\,\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{abc}}{32\pi^4}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\,\log\frac{(w-y)^2}{(x-y)^2}\right.\\
&\left.-\,\frac{(y-w)^i}{(y-w)^2}\,\log\frac{(w-x)^2}{(x-y)^2}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{bd}(x)\,U^{ce}(y)\,\{\rho^d(x),\rho^e(y)\}\right]
\end{aligned}\tag{3.2}$$

Terms $\propto \log\frac{\vee}{k^+}$:

$$\begin{aligned}
\frac{d^3N_{2\,a}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{hbc}}{32\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\,\frac{(w-z)\cdot(y-z)}{(w-z)^2\,(y-z)^2}\right.\\
&\left.-\,\frac{(y-w)^i}{(y-w)^2}\,\frac{(w-z)\cdot(x-z)}{(w-z)^2\,(x-z)^2}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\right]\\
&-\,\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{hbc}}{32\pi^4}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\,\log\frac{(w-y)^2}{(x-y)^2}-\frac{(y-w)^i}{(y-w)^2}\,\log\frac{(w-x)^2}{(x-y)^2}\right]\times\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\right]
\end{aligned}\tag{3.3}$$

Row II, second part, d^3N_{2b}. No large logarithm (R^i depends on $\vee$ only through $\frac{k^+}{\vee}$). Terms $\propto \log\frac{k^+}{\Lambda}$: none.
Terms $\propto \log\frac{\vee}{k^+}$: none.

3.3 Row III

Row III, d^3N_1. Lines 1 and 4 (your green $\frac{1}{\Lambda}$ brackets) are not logarithms. Together with line 1 of d^3N_2 and line 1 of d^3N_{3a} they cancel exactly (other file). Terms $\propto \log\frac{k^+}{\Lambda}$:

$$\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = &\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\left\{\right.\\
&-\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\\
&-\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\left(-\log\left(\frac{k^+}{\Lambda}\right)\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\\
&+\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\left.\right\}
\end{aligned}\tag{3.4}$$

Terms $\propto \log\frac{\vee}{k^+}$:

$$\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = &\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\left\{\right.\\
&-\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\\
&+\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left(\frac{\vee}{k^+}\right)\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\\
&+\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]
\end{aligned}$$

$$
\begin{aligned}
&- \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+} \int_{x,y,z,x'} \frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\, \log\left(\frac{\vee}{k^+}\right) \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \Bigg\}
\end{aligned} \tag{3.5}
$$

Row III, d^3N_2**.** Line 1 is a $\frac{1}{\Lambda}$ term (see d^3N_1). Your two red lines cancel each other and are left out. Terms $\propto \log\frac{k^+}{\Lambda}$:

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = &- \frac{1}{(2\pi)^3} \int_{w,w'} e^{-ik(w'-w)} \Bigg\{ \\
&+ \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+}\, \log\left(\frac{k^+}{\Lambda}\right) \int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&- \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+}\, \log\left(\frac{k^+}{\Lambda}\right) \int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \Bigg\}
\end{aligned} \tag{3.6}
$$

Terms $\propto \log\frac{\vee}{k^+}$:

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = &- \frac{1}{(2\pi)^3} \int_{w,w'} e^{-ik(w'-w)} \Bigg\{ \\
&- \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+}\, \log\left(\frac{\vee}{k^+}\right) \int_{x,y,z,x'} \frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,(x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+ \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+}\, \log\left(\frac{\vee}{k^+}\right) \int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&- \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+}\, \log\left(\frac{\vee}{k^+}\right) \int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+ \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+} \int_{x,y,z,x'} \log\left(\frac{\vee}{k^+}\right) \frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&- \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+} \int_{x,y,z,x'} \log\left(\frac{\vee}{k^+}\right) \frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+ \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+} \int_{x,y,z,x'} \log\left(\frac{\vee}{k^+}\right) \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&- \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+} \int_{x,y,z,x'} \log\left(\frac{\vee}{k^+}\right) \frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-z)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+ \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+} \int_{x,y,z,x'} \log\left(\frac{\vee}{k^+}\right) \frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&- \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+} \int_{x,y,z,x'} \log\left(\frac{\vee}{k^+}\right) \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \Bigg\}
\end{aligned} \tag{3.7}
$$

Row III, d^3N_{3a}**.** Red: the first logarithmic line, with the edge kernel $\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}$, is not integrable at $z \to w$. Blue: it is replaced by $\mathcal{D}$ (fix F3, `BFKL_final.pdf` Sec. 2.1): $U^{bc}(z) \to U^{bc}(w)$ and the kernel $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)$. Without this change the three-ρ terms do not cancel (Sec. 7). Terms $\propto \log\frac{k^+}{\Lambda}$:

$$
\frac{d^3N_{3a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3} \int_{w,w'} e^{-ik(w'-w)} \Bigg\{
$$

$$
\begin{aligned}
&-\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&-\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-w)\,(x-z)\cdot(w-z)}{(x'-w')^2\,(y-w)^2\,(x-z)^2\,(w-z)^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(w)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&-\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&+\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\Bigg\}
\end{aligned}
\tag{3.8}
$$

Terms $\propto \log\frac{\vee}{k^+}$: none.

Row III, d^3N_{3b}**.** No large logarithm. ($\mathcal{A}$ and the logarithms of d^3N_{3a}, d^3N_{3b} have finite limits for $\Lambda\to 0$; other file.) Terms $\propto \log\frac{k^+}{\Lambda}$: none.
Terms $\propto \log\frac{\vee}{k^+}$: none.

3.4 Row IV

Row IV. Terms $\propto \log\frac{k^+}{\Lambda}$:

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik\cdot(w'-w)}\Bigg\{
&+\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left(\frac{k^+}{\Lambda}\right)\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\left[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{e}(y)\rho^{c'}(x)-f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{e}(y)\rho^{d}(x)\right]
\end{aligned}
\tag{3.9}
$$

Terms $\propto \log\frac{\vee}{k^+}$:

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik\cdot(w'-w)}\Bigg\{
&\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left(\frac{\vee}{k^+}\right)\times\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.\\
&\left.-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}+\frac{\delta^{ij}}{2\,(z-w)^2}\right]\\
&-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\times\log\left(\frac{\vee}{k^+}\right)\\
&+\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left(\frac{\vee}{k^+}\right)\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\left[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{e}(y)\rho^{c'}(x)-f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{e}(y)\rho^{d}(x)\right]
\end{aligned}
\tag{3.10}
$$

3.5 Row V

Row V, first part. Terms $\propto \log\frac{k^+}{\Lambda}$:

$$
\begin{aligned}
\frac{d^3N_{1\,\rm NLO}}{d^3k}=&-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y'w'}e^{-ik\cdot(w'-z)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\frac{(x'-w')\cdot(y-z)\,(z-x)\cdot(y'-x)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^{e}(y),\rho^{b}(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^{d}(y),\rho^{e}(y')\}\right]\\
&-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y'w'}e^{-ik\cdot(w'-x)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\\
&\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^{e}(y),\rho^{b}(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^{d}(y),\rho^{e}(y')\}\right]
\end{aligned}
\tag{3.11}
$$

Terms $\propto \log\frac{\vee}{k^+}$: none.

Row V, second part, d^3N_{2a}. Terms $\propto \log \frac{k^+}{\Lambda}$:

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-z)^2}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}\right.$$
$$\left.+\,\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\Big]\,\rho^{b'}(x')\,\rho^f(y)$$
$$-\,\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-x)^2}$$
$$\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\Big]\,\rho^{b'}(x')\,\rho^f(y) \tag{3.12}$$

Terms $\propto \log \frac{\vee}{k^+}$: none.

Row V, second part, $d^3N_{2b\,\mathrm{UV}}$. **E1.** Your bracket $\left[\log\frac{k^+}{\Lambda}+2\log(1-\frac{\Lambda}{k^+})-\frac{11}{6}\right]$ must be $\left[2\log\frac{k^+}{\Lambda}+2\log(1-\frac{\Lambda}{k^+})-\frac{11}{6}\right]$. The error enters in the equation after "Setting $\Lambda\to 0$": there your blue bracket $\left[\log\frac{\vee}{k^+}+\log\frac{k^+}{\Lambda}\right]$ replaces $2\log\frac{\vee}{\Lambda}$ of the equation before, so it needs a factor 2; then $2\log\frac{\vee}{\Lambda}-2\log\frac{\vee}{k^+}=2\log\frac{k^+}{\Lambda}$. (The counterterm, line 6 of $d^3N_{2b\,\mathrm{finite}}$, has $2\log\frac{k^+-\Lambda}{\Lambda}$, as it should.) With the 2 the poles of the $\log\frac{k^+}{\Lambda}$ terms cancel: Group I Row II : Row V 2b UV : Group II Row IV $= 1:-2:1$, sum 8.2×10^{-16} (relative); with your coefficient the sum is 0.5 (`pole_check.py`). Terms $\propto \log\frac{k^+}{\Lambda}$:

$$\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}2\,\log\left(\frac{k^+}{\Lambda}\right)\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}$$
$$\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[U^{af}(y)-U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y). \tag{3.13}$$

Terms $\propto \log\frac{\vee}{k^+}$: none.

Row V, second part, $d^3N_{2b\,\mathrm{finite}}$. Lines 4–6 carry $\log\frac{k^+-\Lambda}{\Lambda}=\log\frac{k^+}{\Lambda}+\log(1-\frac{\Lambda}{k^+})$. Terms $\propto\log\frac{k^+}{\Lambda}$:

$$\frac{d^3N_{2b\,\mathrm{finite\,NLO}}}{d^3k} = -\,\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\left[\frac{(y-x)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right.$$
$$\left.-\,\frac{(y-x)\cdot(x'-w')\,(y-z)\cdot(z-x)}{2(y-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right]\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)$$
$$-\,\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{(y-z)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right.$$
$$\left.+\,\frac{(y-x)\cdot(z-x)\,(y-z)\cdot(x'-w')}{2(y-x)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right]\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)$$
$$-\,\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,2\,\log\left(\frac{k^+}{\Lambda}\right)$$
$$\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y) \tag{3.14}$$

Terms $\propto\log\frac{\vee}{k^+}$: none.

4 Group II

4.1 Row I

Row I, first part, first line. Terms $\propto\log\frac{k^+}{\Lambda}$:

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{$$
$$-\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}$$
$$\Big[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\Big]\times\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big] \tag{4.1}$$

Terms $\propto\log\frac{\vee}{k^+}$:

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{$$

$$
\begin{aligned}
&- \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\log\Big(\frac{\vee}{k^+}\Big)\times\\
&\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]\\
&+\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\\
&-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\\
&\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{4.2}
$$

Row I, first part, second line. Terms $\propto \log\frac{k^+}{\Lambda}$:

$$
\begin{aligned}
\frac{d^3N}{d^3k}=&\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik(w'-w)}\Bigg\{\\
&+\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right]\Bigg\}\\
&\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right]
\end{aligned}
\tag{4.3}
$$

Terms $\propto \log\frac{\vee}{k^+}$:

$$
\begin{aligned}
\frac{d^3N}{d^3k}=&\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik(w'-w)}\Bigg\{\\
&\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\log\Big(\frac{\vee}{k^+}\Big)\times\\
&\left[\frac{\delta^{ij}}{2\,(z-w')^2}+\frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2}-\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right]\\
&-\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w)\cdot(y-z)}{(x'-w)^2\,(y-z)^2\,(z-w')^2}\\
&+\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right]\Bigg\}\\
&\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right]
\end{aligned}
\tag{4.4}
$$

Row I, second part, d^3N_{2a}. No large logarithm: $\vee$ cancels between numerator and denominator of the first logarithm, and the second has the finite limit $\log\frac{(x-w)^2}{(x'-w')^2}$. Terms $\propto \log\frac{k^+}{\Lambda}$: none.
Terms $\propto \log\frac{\vee}{k^+}$: none.

Row I, second part, d^3N_{2b}. Terms $\propto \log\frac{k^+}{\Lambda}$:

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k}=&\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times\\
&\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.\\
&\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\left[(x-z)^2-\frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\right]\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\Big]
\end{aligned}
\tag{4.5}
$$

Terms $\propto \log\frac{\vee}{k^+}$:

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k}=&\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times\\
&\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.
\end{aligned}
\tag{4.6}
$$

$$+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\Bigg]\times\left[(x-z)^2-\frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\right]\times$$
$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)$$
$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]$$
$$-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times$$
$$\Bigg[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}$$
$$+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\Bigg]\times\log\Big(\frac{\vee}{k^+}\Big)\times$$
$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)$$
$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]$$
$$+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^4\,(x-z)^2\,-(x-w)^2\,(x'-z)^2\,(x'-w')^2]}\times$$
$$\Bigg[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}$$
$$+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\Bigg]\times\log\Big(\frac{\vee}{k^+}\Big)$$
$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)$$
$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\tag{4.6}$$

Row I, second part, d^3N_{2c}. Terms $\propto \log\frac{k^+}{\Lambda}$: none.
Terms $\propto \log\frac{\vee}{k^+}$:

$$\frac{d^3N_{2c\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-w')^2}{(x'-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}$$
$$\Bigg[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}$$
$$+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\Bigg]\times\log\Big(\frac{\vee}{k^+}\Big)$$
$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)$$
$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]$$
$$-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-w)^2}{(x-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}$$
$$\Bigg[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}$$
$$+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\Bigg]\times\log\Big(\frac{\vee}{k^+}\Big)$$
$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)$$
$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\tag{4.7}$$

Row I, second part, d^3N_{2d}. Terms $\propto \log\frac{k^+}{\Lambda}$: none.
Terms $\propto \log\frac{\vee}{k^+}$:

$$\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k}=-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]}$$

$$
\Bigg[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}
$$

$$
+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\Bigg]
$$

$$
\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$

$$
-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]
$$

$$
+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\Bigg[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}
$$

$$
+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\Bigg]\times
$$

$$
\log\left(\frac{\vee}{k^+}\right)\times
$$

$$
\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$

$$
-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]
$$

$$
+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\Bigg[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}
$$

$$
+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\Bigg]\times
$$

$$
\log\left(\frac{\vee}{k^+}\right)\times
$$

$$
\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$

$$
-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg] \tag{4.8}
$$

Row I, second part, d^3N_{2e}. Terms $\propto \log\frac{k^+}{\Lambda}$: none.
Terms $\propto \log\frac{\vee}{k^+}$:

$$
\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]}
$$

$$
\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}
$$

$$
+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]
$$

$$
\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$

$$
-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]
$$

$$
-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-w)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}
$$

$$
+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\times
$$

$$
\log\left(\frac{\vee}{k^+}\right)\times
$$

$$
\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$

$$
\begin{aligned}
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big] \\
&- \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-w')^2}{[(x'-w')^2-(x'-z)^2]}\, \frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
&\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2} \\
&+ \frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\times \\
&\log\left(\frac{\vee}{k^+}\right)\times \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big]
\end{aligned}
\tag{4.9}
$$

Row I, second part, d^3N_{2f}. Terms $\propto \log\frac{k^+}{\Lambda}$: none.
Terms $\propto \log\frac{\vee}{k^+}$:

$$
\begin{aligned}
\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\, \frac{g^4\, d_\perp}{32\pi^5}\, \log\left(\frac{\vee}{k^+}\right) \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{\left[(x'-w')^2\,(x-w)^2-(x-z)^2\,(x'-z)^2\right]}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]\left[(x'-z)^2-(x'-w')^2\right]} \\
&\Big[f^{c'bd'}\, U^{ac'}(w')\, \rho^{d'}(x') - f^{ac'd'}\, U^{bc'}(z) U^{e'd'}(x')\, \rho^{e'}(x')\Big]\times \Big[f^{cbd}\, U^{ac}(w)\, \rho^{d}(x) - f^{acd}\, U^{bc}(z)\, U^{de}(x)\, \rho^{e}(x)\Big] \\
&- \frac{1}{(2\pi)^3}\, \frac{g^4\, d_\perp}{32\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x-z)^2\,(x-w)^2}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]}\, \frac{\left[(x'-z)^2-(x'-w')^2\right]}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\times \\
&\log\left(\frac{\vee}{k^+}\right)\times \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big] \\
&+ \frac{1}{(2\pi)^3}\, \frac{g^4\, d_\perp}{32\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-w')^2\,(x'-z)^2}{(z-w)^2\,(z-w')^2} \\
&\frac{\left[(x-z)^2-(x-w)^2\right]}{\left[(x'-z)^2-(x'-w')^2\right]\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\times \log\left(\frac{\vee}{k^+}\right)\times \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big]
\end{aligned}
\tag{4.10}
$$

4.2 Row II

Only four ρ's: Sec. 6 (with the same coefficient for both logarithms).

4.3 Row III

Only four ρ's: Sec. 6 (with the same coefficient for both logarithms).

4.4 Row IV

Row IV. Hidden logarithm. In line 2, $\left[\log\frac{k^+}{\Lambda} + \log(\frac{\vee}{k^+} - 1)\right]$, the $\log\frac{\vee}{k^+}$ is cancelled by the one in the $\frac{\xi}{(1-\xi)_+}$ integral of line 1 (lower limit $\frac{k^+}{\vee}$). It is not a BFKL term. The BFKL term of this row is $\log\frac{\vee}{k^+} H_4$ (blue): at $\xi \to \frac{k^+}{\vee}$ the phase $e^{-ik\cdot(y'-y)\frac{1}{\xi}}$ probes $|y'-y| \to 0$, and the large-momentum tail of $\log(y'-y)^2$ gives $y, y' \to z$ (`Finite_terms_final.pdf`, C1).
Terms $\propto \log\frac{k^+}{\Lambda}$:

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} &= +\frac{g^2}{4\pi^2}\, \frac{1}{(2\pi)^3}\, \frac{g^2}{2\pi^2}\, \frac{N_c}{k^+}\, \log\left(\frac{k^+}{\Lambda}\right) \int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\, \frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\, \Bigg[-\frac{d_\perp}{\epsilon} - d_\perp\, \gamma_E \\
&- \frac{d_\perp}{2}\, \log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right) - 1\Bigg]\, \left[U^{e'c}(y') - U^{e'c}(x')\right]\times \left[U^{ec}(y) - U^{ce}(x)\right]\, \rho^{e'}(x')\rho^{e}(x)
\end{aligned}
\tag{4.11}
$$

Terms $\propto \log\frac{\vee}{k^+}$:

$$
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\, \frac{g^4\, N_c}{8\pi^5}\, \frac{1}{k^+}\, \log\left(\frac{\vee}{k^+}\right) \int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\, \frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\, \frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}
$$

$$\left[U^{e'c}(z)-U^{e'c}(x')\right]\times\left[U^{ec}(z)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x) \tag{4.12}$$

5 Group III (with c.c.)

5.1 Row I

Row I. Terms $\propto \log\frac{k^+}{\Lambda}$:

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k}=&\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'}e^{-ik(w'-w)}\Bigg\{\\&-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\left[\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}+\frac{(x-z)^j(x-w)^i}{2(x-z)^2(x-w)^2}\right]\Bigg\}\times\\&\left[U^{ab'}(x')U^{c'd'}(y)U^{c'b}(z)\rho^{b'}(x')\rho^{d'}(y)-U^{bd'}(z)U^{d'e'}(y)U^{ac'}(w)\rho^{c'}(x')\rho^{e'}(y)\right]\times\\&\left[f^{cbd}U^{ac}(w)\rho^d(x)-f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)\right]\end{aligned} \tag{5.1}$$

Terms $\propto \log\frac{\vee}{k^+}$:

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k}=&\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'}e^{-ik(w'-w)}\Bigg\{\\&-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\times\log\left(\frac{\vee}{k^+}\right)\times\left[\frac{\delta^{ij}}{2(z-w)^2}+\frac{(x-z)^j(z-w)^i}{(z-w)^2(x-z)^2}\right.\\&\left.-\frac{(x-z)^j(x-w)^i}{(x-w)^2(x-z)^2}-\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}\right]\\&+\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}\\&-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\left[\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}+\frac{(x-z)^j(x-w)^i}{2(x-z)^2(x-w)^2}\right]\Bigg\}\times\\&\left[U^{ab'}(x')U^{c'd'}(y)U^{c'b}(z)\rho^{b'}(x')\rho^{d'}(y)-U^{bd'}(z)U^{d'e'}(y)U^{ac'}(w)\rho^{c'}(x')\rho^{e'}(y)\right]\times\\&\left[f^{cbd}U^{ac}(w)\rho^d(x)-f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)\right]\end{aligned} \tag{5.2}$$

5.2 Row II

Row II. Terms $\propto \log\frac{k^+}{\Lambda}$:

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k}=&\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'}e^{-ik(w'-w)}\Bigg\{\\&-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\left[\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}+\frac{(x-z)^j(x-w)^i}{2(x-z)^2(x-w)^2}\right]\Bigg\}\times\\&\left[U^{c'd'}(y)U^{ab'}(x')U^{bc'}(z)\rho^{d'}(y)\rho^{b'}(x')-U^{ab'}(x')\rho^b(y)\rho^{b'}(x')\right]\times\left[f^{cbd}U^{ac}(w)\rho^d(x)-f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)\right]\end{aligned} \tag{5.3}$$

Terms $\propto \log\frac{\vee}{k^+}$:

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k}=&\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'}e^{-ik(w'-w)}\Bigg\{\\&-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\times\log\left(\frac{\vee}{k^+}\right)\times\left[\frac{\delta^{ij}}{2(z-w)^2}+\frac{(x-z)^j(z-w)^i}{(z-w)^2(x-z)^2}\right.\\&\left.-\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}-\frac{(x-z)^j(x-w)^i}{(x-z)^2(x-w)^2}\right]\\&+\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}\\&-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\left[\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}+\frac{(x-z)^j(x-w)^i}{2(x-z)^2(x-w)^2}\right]\Bigg\}\times\\&\left[U^{c'd'}(y)U^{ab'}(x')U^{bc'}(z)\rho^{d'}(y)\rho^{b'}(x')-U^{ab'}(x')\rho^b(y)\rho^{b'}(x')\right]\times\left[f^{cbd}U^{ac}(w)\rho^d(x)-f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)\right]\end{aligned} \tag{5.4}$$

5.3 Row III

Row III, first part. Hidden logarithm. The $\log \frac{\vee}{k^+}$ in $\left[\log \frac{k^+}{\Lambda} + \log(\frac{\vee}{k^+} - 1)\right]$ is cancelled by the one in the $\left[\frac{1}{p^+}\right]_+$ integral of line 1 (its subtraction $-\frac{f(0)}{p^+}$ up to $\vee - k^+$). It is not a BFKL term. The BFKL term of the row is the blue term (B) of `BFKL_final.pdf`: the terms of the p^+ bracket that tend to a constant at large p^+ give $\log \frac{\vee}{k^+}$ after $\int_{y'}$, with $y' \to z$. Terms $\propto \log \frac{k^+}{\Lambda}$:

$$\frac{d^3 N_{1\,\mathrm{NLO}}}{d^3 k} = -\frac{1}{(2\pi)^3} \frac{i g^4 f^{c'd'a}}{16\pi^5} \frac{1}{k^+} \log\left(\frac{k^+}{\Lambda}\right) \int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)} \frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right] \times \left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right] \quad (5.5)$$

Terms $\propto \log \frac{\vee}{k^+}$:

$$\frac{d^3 N_{1\,\mathrm{NLO}}}{d^3 k} = -\frac{1}{(2\pi)^3} \frac{i g^4 f^{c'd'a}}{16\pi^5} \frac{1}{k^+} \log\left(\frac{\vee}{k^+}\right) \int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)} \frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]$$
$$\times \left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right] \quad (5.6)$$

Row III, second part. Hidden logarithm as in the first part; the blue term is (B). **C5**: the measure is $\int_{w,x',y',z,x}$ (the row has $\rho(x')\rho(x)$; there is no y and no w'). Terms $\propto \log \frac{k^+}{\Lambda}$:

$$\frac{d^3 N_{2\,\mathrm{NLO}}}{d^3 k} = -\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^5} \frac{1}{k^+} \log\left(\frac{k^+}{\Lambda}\right) \int_{w,y',z,y,x} \int_{w,x',y',z,x} e^{-ik\cdot(y'-w)} \left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right.$$
$$\left. + \frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right] \left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right] \times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \quad (5.7)$$

Terms $\propto \log \frac{\vee}{k^+}$:

$$\frac{d^3 N_{2\,\mathrm{NLO}}}{d^3 k} = -\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^5} \frac{1}{k^+} \log\left(\frac{\vee}{k^+}\right) \int_{w,x',y',z,x} e^{-ik\cdot(y'-w)} \frac{(x'-z)\cdot(x-z)}{(x'-z)^2\,(x-z)^2}$$
$$\times \left[\frac{(y'-z)\cdot(z-w)}{(y'-z)^2\,(z-w)^2} - \frac{(y'-z)\cdot(x-w)}{2\,(y'-z)^2\,(x-w)^2}\right]$$
$$\left[f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]$$
$$\times \left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \quad (5.8)$$

5.4 Row IV

Only four ρ's: Sec. 6 (with the same coefficient for both logarithms).

5.5 Row V

Row V. Hidden logarithm as in Row III; the blue term is (B). Terms $\propto \log \frac{k^+}{\Lambda}$:

$$\frac{d^3 N_{\mathrm{NLO}}}{d^3 k} = -\frac{1}{(2\pi)^3} \frac{i g^4 f^{c'ca}}{8\pi^5} \frac{1}{k^+} \log\left(\frac{k^+}{\Lambda}\right) \int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)} \frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\right] \times \left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right] \quad (5.9)$$

Terms $\propto \log \frac{\vee}{k^+}$:

$$\frac{d^3 N_{\mathrm{NLO}}}{d^3 k} = -\frac{1}{(2\pi)^3} \frac{i g^4 f^{c'ca}}{8\pi^5} \frac{1}{k^+} \log\left(\frac{\vee}{k^+}\right) \int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)} \frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(z)\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\right] \left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right] \quad (5.10)$$

5.6 Row VI

Row VI. Hidden logarithm as in Row III; the blue term is (B). Terms $\propto \log \frac{k^+}{\Lambda}$:

$$\frac{d^3 N_{\mathrm{NLO}}}{d^3 k} = -\frac{1}{(2\pi)^3} \frac{i g^4 f^{c'd'a}}{8\pi^5} \frac{1}{k^+} \log\left(\frac{k^+}{\Lambda}\right) \int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)} \frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right] \times \left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\right] \quad (5.11)$$

Terms $\propto \log \frac{\vee}{k^+}$:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]$$
$$\times\,\Big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x) - U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\Big] \tag{5.12}$$

6 Four ρ's

Group I Row I, Group II Rows II and III and Group III Row IV have only four ρ's. They are your `4rho.tex`, Eq. (4rho), with

$$\log\frac{\vee}{\Lambda} = \log\frac{k^+}{\Lambda} + \log\frac{\vee}{k^+}\,:$$

every four-ρ term has the same coefficient for both logarithms.

- The four-ρ terms cancel after symmetrization: your demonstration in `4rho.tex` (two and four Wilson lines; three Wilson lines by $x \leftrightarrow y$); checked numerically before (4.7×10^{-15}, `Final_evolution_check.pdf`).
- Their three-ρ remainders, your final forms, enter both three-ρ lists of Sec. 7:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{i}{(2\pi)^3}\,\frac{g^4}{8\pi^3}\,\frac{\Big[\log\Big(\frac{k^+}{\Lambda}\Big)+\log\Big(\frac{\vee}{k^+}\Big)\Big]}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Big[$$
$$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^{d}(x')\rho^{a}(y)\rho^{c}(x)\,U^{de}(w')U^{eb}(w)$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^{e}(x')\rho^{a}(y)\rho^{c}(x)\,U^{db}(w)U^{ed}(x')$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^{c}(x')\rho^{a}(y)\rho^{e}(x)\,U^{db}(w')U^{ed}(x)$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^{c}(x')\rho^{a}(y)\rho^{e}(x)U^{bd}(w')U^{ed}(w)\Big] \tag{6.1}$$

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{i}{(2\pi)^3}\,\frac{g^4}{8\pi^3}\,\frac{\Big[\log\Big(\frac{k^+}{\Lambda}\Big)+\log\Big(\frac{\vee}{k^+}\Big)\Big]}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Big[$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^{d}(x')\rho^{h}(y)\rho^{e}(x)U^{db}(w')U^{ec}(x)U^{ah}(y)$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^{d}(x')\rho^{e}(x)\rho^{h}(y)U^{dc}(x')U^{eb}(w)U^{ah}(y)$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^{d}(x')\rho^{e}(x)\rho^{h}(y)U^{dc}(x')U^{be}(x)U^{ah}(y)$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^{e}(x')\rho^{h}(y)\rho^{d}(x)U^{dc}(x)U^{be}(x')U^{ah}(y)\Big] \tag{6.2}$$

- Their two-ρ remainders (your "Two ρ contribution", with F1, F2) enter both two-ρ sums, Secs. 8 and 9.

7 The three-ρ terms cancel

All terms with three ρ's of Secs. 3–6, for each logarithm separately. Which terms have three ρ's is decided numerically ($\rho \to 2\rho$ multiplies them by 8).

Method.

- The charges are integration variables, so only the totally symmetric part of $\rho\rho\rho$ can cancel against other terms. The rest (commutators, $[\rho^a(u),\rho^b(v)] = if^{abc}\rho^c(u)\delta(u-v)$) are two-$\rho$ terms; they are in Secs. 8 and 9. The symmetric part is evaluated with classical charges.
- In every term the integration variables are renamed so that the phase is $e^{-ik\cdot(w'-w)}$, the soft gluon is z and the charges are x, y, x': in Group I Row V part 1 the measured gluon of the amplitude is at z (line 3) or x (line 4); in Group III Rows III, V, VI and in (B) the conjugate one is at y'.
- Each term is summed over the 6 placements of its three charges and, for Groups I and III, its c.c. is added (conjugate, $w \leftrightarrow w'$). Group II Row I is self-conjugate as written: its second line is the c.c. of the first. The `4rho.tex` lists are self-conjugate as written.
- In Group I Row II d^3N_{2a}, lines 2 and 3, the z integral is done. For a pointwise test it is written back: $\log\frac{(w-y)^2}{(x-y)^2} = -\frac{1}{\pi}\int_z \big[K(w,y,z) - K(x,y,z)\big]$, $K(a,b,z) = \mathcal{K}(a-z)\cdot\mathcal{K}(b-z)$.
- Random points, random SU(3) adjoint Wilson lines with $U^{bc} = U^{cb}$, random charges; three configurations.

Values at one configuration, in units of $i\mathcal{N}\,e^{-ik\cdot(w'-w)}$, with the large logarithm set to 1 (all values are real in these units):

Three ρ's $\propto \log \frac{k^+}{\Lambda}$ (22 terms).

term		direct	c.c.	sum
Group I II(2a), line 1	Eq. (3.2)	-0.8549	-14.6946	-15.5494
Group I II(2a), line 3	Eq. (3.2)	$+1.0769$	$+0.2120$	$+1.2889$
Group I III d^3N_1, line 2	Eq. (3.4)	$+45.4456$	$+0.0289$	$+45.4745$
Group I III d^3N_1, line 5	Eq. (3.4)	-45.4456	-0.0289	-45.4745
Group I III d^3N_1, line 7	Eq. (3.4)	-0.4492	-1.2495	-1.6987
Group I III d^3N_2, line 5	Eq. (3.6)	-19.8506	$+0.1921$	-19.6585
Group I III d^3N_2, line 6	Eq. (3.6)	-30.9162	-2.0644	-32.9806
Group I III d^3N_{3a}, line 2 ($\mathcal{D}$)	Eq. (3.8)	-15.7868	-25.4825	-41.2693
Group I III d^3N_{3a}, line 3	Eq. (3.8)	$+30.9162$	$+2.0644$	$+32.9806$
Group I III d^3N_{3a}, line 4	Eq. (3.8)	$+19.8506$	-0.1921	$+19.6585$
Group I IV, line 3	Eq. (3.9)	-0.0717	-9.8576	-9.9294
Group I V(1), line 3	Eq. (3.11)	-0.2873	-0.4411	-0.7284
Group I V(1), line 4	Eq. (3.11)	-0.2873	-0.4411	-0.7284
Group II I(1) A, line 3	Eq. (4.1)	-1.8952	(self-conj.)	-1.8952
Group II I(1) B, line 3	Eq. (4.3)	$+5.2217$	(self-conj.)	$+5.2217$
Group III I, line 3	Eq. (5.1)	$+0.0717$	$+9.8576$	$+9.9294$
Group III II, line 3	Eq. (5.3)	$+1.5672$	$+3.4600$	$+5.0271$
Group III III(1), line 2	Eq. (5.5)	-0.1498	$+0.5177$	$+0.3679$
Group III V, line 2	Eq. (5.9)	$+25.4825$	$+15.7868$	$+41.2693$
Group III VI, line 2	Eq. (5.11)	$+2.6061$	$+1.5813$	$+4.1874$
`4rho.tex`, two Wilson lines	Eq. (6.1)	$+6.3001$	(in the term)	$+6.3001$
`4rho.tex`, three Wilson lines	Eq. (6.2)	-1.7932	(in the term)	-1.7932
total				$-3.5e-14$

Largest relative residual over three configurations: as in the note: 7.7×10^{-16}; as in the note, 4rho.tex lists with an extra c.c.: 9.9×10^{-2}; edge kernel instead of D: 5.9×10^{-1}; edge kernel instead of D, 4rho.tex lists with an extra c.c.: 6.9×10^{-1}.

Three ρ's $\propto \log \frac{\vee}{k+}$ (35 terms).

term		direct	c.c.	sum
Group I II(2a), line 1	Eq. (3.3)	-0.8549	-14.6946	-15.5494
Group I II(2a), line 2	Eq. (3.3)	-0.5232	$+3.4724$	$+2.9492$
Group I III d^3N_1, line 2	Eq. (3.5)	$+45.4456$	$+0.0289$	$+45.4745$
Group I III d^3N_1, line 3	Eq. (3.5)	-45.4456	-0.0289	-45.4745
Group I III d^3N_1, line 7	Eq. (3.5)	-0.4492	-1.2495	-1.6987
Group I III d^3N_1, line 8	Eq. (3.5)	$+0.8984$	$+2.4989$	$+3.3973$
Group I III d^3N_2, line 2	Eq. (3.7)	$+1.5244$	$+1.5741$	$+3.0986$
Group I III d^3N_2, line 5	Eq. (3.7)	-19.8506	$+0.1921$	-19.6585
Group I III d^3N_2, line 6	Eq. (3.7)	-30.9162	-2.0644	-32.9806
Group I III d^3N_2, line 7	Eq. (3.7)	$+4.9673$	-2.0824	$+2.8850$
Group I III d^3N_2, line 8	Eq. (3.7)	$+3.0331$	-0.7494	$+2.2837$
Group I III d^3N_2, line 9	Eq. (3.7)	$+30.9162$	$+2.0644$	$+32.9806$
Group I III d^3N_2, line 10	Eq. (3.7)	$+9.5357$	$+0.0611$	$+9.5968$
Group I III d^3N_2, line 11	Eq. (3.7)	-21.2938	-0.2070	-21.5009
Group I III d^3N_2, line 12	Eq. (3.7)	$+19.8506$	-0.1921	$+19.6585$
Group I IV, line 1	Eq. (3.10)	$+3.4365$	$+1.9039$	$+5.3403$
Group I IV, line 2	Eq. (3.10)	-0.5655	-1.2722	-1.8377
Group I IV, line 3	Eq. (3.10)	-0.0717	-9.8576	-9.9294
Group II I(1) A, line 1	Eq. (4.2)	$+1.9040$	(self-conj.)	$+1.9040$
Group II I(1) A, line 2	Eq. (4.2)	$+1.5852$	(self-conj.)	$+1.5852$
Group II I(1) A, line 3	Eq. (4.2)	-1.8952	(self-conj.)	-1.8952
Group II I(1) B, line 1	Eq. (4.4)	-2.0102	(self-conj.)	-2.0102
Group II I(1) B, line 2	Eq. (4.4)	$+0.0433$	(self-conj.)	$+0.0433$
Group II I(1) B, line 3	Eq. (4.4)	$+5.2217$	(self-conj.)	$+5.2217$
Group III I, line 1	Eq. (5.2)	-3.4365	-1.9039	-5.3403
Group III I, line 2	Eq. (5.2)	$+0.5655$	$+1.2722$	$+1.8377$
Group III I, line 3	Eq. (5.2)	$+0.0717$	$+9.8576$	$+9.9294$
Group III II, line 1	Eq. (5.4)	-3.1195	-1.3034	-4.4229
Group III II, line 2	Eq. (5.4)	$+4.0897$	$+0.4887$	$+4.5784$
Group III II, line 3	Eq. (5.4)	$+1.5672$	$+3.4600$	$+5.0271$
(B), Group III Row III(1)	Eq. (5.6)	-1.4036	-2.2332	-3.6368

(B), Group III Row V	Eq. (5.10)	$+1.4036$	$+2.2332$	$+3.6368$
(B), Group III Row VI	Eq. (5.12)	-0.0000	$+0.0000$	-0.0000
`4rho.tex`, two Wilson lines	Eq. (6.1)	$+6.3001$	(in the term)	$+6.3001$
`4rho.tex`, three Wilson lines	Eq. (6.2)	-1.7932	(in the term)	-1.7932
total				$+4.9e-15$

Largest relative residual over three configurations: as in the note: 1.5×10^{-16}; as in the note, 4rho.tex lists with an extra c.c.: 9.9×10^{-2}; without the terms (B): 3.1×10^{-16}; without the terms (B), 4rho.tex lists with an extra c.c.: 9.9×10^{-2}.

So:

- **The three-ρ terms $\propto \log \frac{k^+}{\Lambda}$ cancel**, with $\mathcal{D}$ in place of the edge-kernel term of Group I Row III d^3N_{3a}. With the edge kernel they do not.
- **The three-ρ terms $\propto \log \frac{\vee}{k^+}$ cancel.** The (B) terms of Group III Rows III(1) and V cancel each other, that of Row VI vanishes by itself after symmetrization; so the three-ρ check does not test (B). (B) is tested by the two-ρ sum, Sec. 9.
- Adding a c.c. to the `4rho.tex` lists would double them: they already contain it.
- Group I Row IV and Group III Row I cancel each other line by line, and so do several pairs of lines in Group I Row III.

8 The two-ρ terms $\propto \log \frac{k^+}{\Lambda}$: JIMWLK

Number of terms (green / purple): genuine, expanded: 135 / 4; `4rho.tex` lists: 0 / 30; from reordering three and four ρ's: 208 / 127. No orange term appears.

8.1 Genuine two-ρ terms of the rows

These are the two-ρ terms $\propto \log \frac{k^+}{\Lambda}$ of Secs. 3–5 (Group I Rows II and V, Group II Rows I and IV, Group III Row III part 2), written in a common notation. To compare term by term, every dimensionally regulated bracket is written back as the z integral it comes from (exact in $d_\perp = 2-\epsilon$, $\overline{\text{MS}}$):

- Group I Row II: $\frac{2}{\epsilon} + 2\gamma + \log\frac{(x-w)^2\mu^2}{4} = -\frac{1}{\pi}\int_z \big[K(x,w,z) - K(w,w,z)\big]$;
- Group I Row V 2b UV: $\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu^2}{4} = \frac{1}{\pi}\int_x \big[K(z,z,x) - K(y,z,x)\big]$;
- Group II Row IV: $-\frac{d_\perp}{\epsilon} - d_\perp\gamma_E - \frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2}{4} - 1 = \frac{1}{\pi}\int_z \big[K(y,y',z) - \frac{1}{2}K(y,y,z) - \frac{1}{2}K(y',y',z)\big] + O(\epsilon)$.

The terms are expanded, the charges are named x (emission ending at w) and x' (ending at w'), and the colour algebra is done. Group I and Group III include their c.c.

- **Row V 2b: the UV term and the finite counterterm cancel each other.** Both have the kernel $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,[K(z,z,x)-K(y,z,x)]$ and the colour $N_c[U(y)-U(z)]$.
  - UV: $+\frac{g^4N_c}{8\pi^4}\cdot\frac{1}{\pi}\int_x$.
  - Counterterm: $-\frac{g^4}{8\pi^5}N_c$, pointwise in x.

  So the $\log\frac{k^+}{\Lambda}$ part of Row V 2b is just its two endpoint terms. This is the same as before the UV subtraction.
- Row V 2b UV is taken with the coefficient $2\log\frac{k^+}{\Lambda}$ (E1).

Group I, Row II.

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|_{G1\text{-}II} = \mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)
\end{aligned}
$$

$$
\begin{aligned}
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Big]
\end{aligned}
\tag{8.1}
$$

Group I, Row V, part 2a.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G1\text{-}V\ 2a} &= \mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(x)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(w)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{al}(x')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{al}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\Bigg]
\end{aligned}
\tag{8.2}
$$

Group I, Row V, part 2b finite: the two endpoint terms (B_2, B_4).

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G1\text{-}V\ 2b\,\mathrm{end}} &= \mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)
\end{aligned}
$$

$$
\begin{aligned}
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\Big]
\end{aligned}
\tag{8.3}
$$

Group I, Row V, part 2b UV (bracket written as $\frac{1}{\pi}\int_x[K(z,z,x)-K(y,z,x)]$).

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G1\text{-}V\ 2b\ \mathrm{UV}} &= \mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Big]
\end{aligned}
\tag{8.4}
$$

Group I, Row V, part 2b finite: the counterterm.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G1\text{-}V\ 2b\ \mathrm{CT}} &= \mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)
\end{aligned}
$$

$$
\begin{aligned}
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Big]
\end{aligned}
\tag{8.5}
$$

Group II, Row I.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G2\text{-}I} = \mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[&\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\Big]
\end{aligned}
\tag{8.6}
$$

Group II, Row IV.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G2\text{-}IV} = \mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[&\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w')\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)\Big]
\end{aligned}
\tag{8.7}
$$

Group III, Row III, second part.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G3\text{-}III(2)} = \mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[&\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)
\end{aligned}
$$

$$
\begin{aligned}
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&-4\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w')\,U^{cb}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{cb}(w)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{la}(w')\,U^{be}(z)\,U^{cd}(w)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{la}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{la}(w')\,U^{be}(z)\,U^{cd}(w)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{la}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^{h}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;U^{ab}(w)\,U^{bc}(x')\,\rho^{c}(x')\,\rho^{a}(x)\Big]
\end{aligned}
\tag{8.8}
$$

8.2 From four ρ's: the two-ρ lists of `4rho.tex`

These are the two-ρ terms of `4rho.tex` ("Two ρ contribution", $dN^{(1)}$, $dN^{(2)}$ with c.c.), with the two fixes found earlier (both still needed in this `4rho.tex`):

- **F1:** in the first part, drop $-8N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,K(y,x',z)$;
- **F2:** in the second part, $dN^{(1)}$, term 2 has one index four times. The correct term is

$$3f^{abc}f^{deg}\rho^{b'}(x')\rho^{c}(y)U^{ag}(y)U^{db'}(w')U^{eb}(z).$$

Collected (added):

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{4\rho\text{-lists}} = \mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(w)\,U^{dh}(x)\,U^{ml}(w')\,\rho^{m}(x')\,\rho^{e}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^{m}(x')\,\rho^{e}(x)\\
&-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(w)\,U^{bl}(x)\,U^{dl}(z)\,U^{mh}(w')\,\rho^{m}(x')\,\rho^{e}(x)\\
&+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{dl}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^{m}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ha}(w)\,U^{hl}(x')\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{ad}(z)\,U^{bh}(x)\,U^{el}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{ah}(x)\,U^{dl}(w')\,U^{eb}(z)\,\rho^{l}(x')\,\rho^{c}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x')\,\rho^{e}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\tfrac{3}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w)\,U^{lh}(x')\,U^{da}(x)\,\rho^l(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{hd}(x)\,\rho^e(x)\,\rho^c(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(x')\,U^{lh}(x)\,U^{da}(z)\,\rho^l(x)\,\rho^c(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{lc}(x')\,U^{be}(x)\,\rho^h(x)\,\rho^l(x')\Big]
\end{aligned}
\tag{8.9}
$$

8.3 From reordering the three and four ρ's

The commutator terms of the three-ρ terms of Sec. 7 and of the four-ρ terms. They are your lists of `Evolution_terms.tex`, Sec. 2, which were checked row by row against these rows (0 differing kernel monomials in every row, `Final_evolution_check.pdf`), here with $\log\frac{\vee}{\Lambda}\to\log\frac{k^+}{\Lambda}$ and coloured.

From $4\,\rho's$

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Big[\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{bc}(w)\,\rho^a(x')\,\rho^c(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w)\,U^{ca}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{ca}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^e(x')\,\rho^c(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{dh}(x')\,\rho^e(x')\,\rho^c(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{dh}(x)\,\rho^c(x')\,\rho^e(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{lc}(x)\,U^{ad}(x')\,\rho^h(x')\,\rho^l(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^l(x')\,\rho^h(x)\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{dh}(w')\,U^{hb}(w)\,\rho^a(x)\,\rho^e(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{dh}(x')\,\rho^a(x)\,\rho^e(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hb}(w')\,U^{dh}(x')\,\rho^a(x)\,\rho^e(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{bh}(w')\,U^{dh}(w)\,\rho^a(x)\,\rho^e(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{dc}(x')\,U^{al}(x)\,\rho^l(x)\,\rho^h(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ec}(x')\,U^{db}(w)\,U^{al}(x)\,\rho^h(x')\,\rho^l(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Big]
\end{aligned}
\tag{8.10}
$$

From three ρ's

Group I (with c.c.)

Row I Only 4 ρ's

Row II

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Big[\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x') \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ah}(w')\,\rho^b(x')\,\rho^e(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,\rho^b(x')\,\rho^e(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x') \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x') \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x') \\
&+2\,N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x) \\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x') \\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^e(x')\,\rho^c(x) \\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^e(x')\,\rho^c(x) \\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x) \\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ah}(w')\,\rho^c(x)\,\rho^e(x') \\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ah}(w')\,\rho^c(x)\,\rho^e(x') \\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')\Big]
\end{aligned}
\tag{8.11}
$$

Row III We write it in two parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Big[\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x') \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)
\end{aligned}
$$

$$
\begin{aligned}
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x)\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x)\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^l(x)\,\rho^h(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')\Big]
\end{aligned}
\tag{8.12}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Big[\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(w)\,\rho^l(x')\,\rho^h(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(w)\,\rho^l(x')\,\rho^h(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(w')\,\rho^l(x')\,\rho^h(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(x')\,\rho^l(x')\,\rho^h(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(w)\,\rho^h(x')\,\rho^l(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(w')\,\rho^h(x')\,\rho^l(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(w')\,\rho^h(x')\,\rho^l(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(w)\,\rho^l(x)\,\rho^h(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(w)\,\rho^l(x)\,\rho^h(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{cl}(x)\,U^{be}(w')\,\rho^l(x)\,\rho^h(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x)\,U^{be}(w')\,\rho^l(x)\,\rho^h(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x)\,U^{be}(x')\,\rho^l(x)\,\rho^h(x')\Big]
\end{aligned}
\tag{8.13}
$$

Row IV We write it in two parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Big[\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ca}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{le}(w')\,U^{al}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{al}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{al}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{be}(x)\,U^{ad}(w)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{le}(w')\,U^{al}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{cb}(x')\,\rho^{c}(x')\,\rho^{a}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bd}(x')\,U^{lc}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ca}(x')\,\rho^{b}(x)\,\rho^{c}(x')\Bigg]
\end{aligned}
\tag{8.14}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ca}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bl}(x)\,U^{dc}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bl}(x)\,U^{ad}(w)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\Bigg]
\end{aligned}
\tag{8.15}
$$

Row V

$$
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[
$$

$$
\begin{aligned}
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{be}(z)\,U^{cl}(w')\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(w')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{bd}(z)\,U^{cl}(w)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(w)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{bl}(z)\,U^{cd}(w)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(w)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{bl}(z)\,U^{ce}(w')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(w')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\Big]
\end{aligned}
\tag{8.16}
$$

Group II

Row I

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
&\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ca}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{be}(z)\,U^{al}(w')\,U^{cd}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{bm}(z)\,U^{md}(x)\,U^{ah}(w)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{bc}(w)\,\rho^{c}(x)\,\rho^{a}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hm}(x)\,U^{ld}(x')\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ce}(x')\,U^{al}(w)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ca}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{eb}(z)\,U^{cd}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{bm}(z)\,U^{md}(x)\,U^{ha}(w)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{bc}(w)\,\rho^{c}(x)\,\rho^{a}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hm}(x)\,U^{ld}(x')\,\rho^{m}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ce}(x')\,U^{db}(z)\,U^{al}(w)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ae}(w')\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{bl}(z)\,U^{lm}(x)\,U^{ah}(w)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{hd}(w)\,\rho^{e}(x')\,\rho^{b}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hd}(x')\,U^{lm}(x)\,\rho^{e}(x')\,\rho^{m}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ce}(x')\,U^{ad}(w)\,\rho^{h}(x')\,\rho^{l}(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{lb}(z)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{bl}(z)\,U^{lm}(x)\,U^{ha}(w)\,\rho^m(x)\,\rho^e(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ca}(x')\,\rho^b(x)\,\rho^c(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{hd}(w)\,\rho^e(x')\,\rho^b(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hd}(x')\,U^{lm}(x)\,\rho^e(x')\,\rho^m(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ce}(x')\,U^{lb}(z)\,U^{ad}(w)\,\rho^h(x')\,\rho^l(x)\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Big]
\end{aligned}
\tag{8.17}
$$

Row II Only 4 ρ's

Row III Only 4 ρ's

Row IV Only 2 ρ's

Group III (with c.c.)

Row I We write it in two parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Big[\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{md}(x)\,U^{mb}(z)\,U^{ah}(w)\,\rho^l(x')\,\rho^e(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hd}(x)\,U^{lm}(w')\,U^{al}(w)\,\rho^m(x')\,\rho^e(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{mb}(z)\,U^{ah}(w')\,\rho^e(x')\,\rho^l(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hd}(x')\,U^{lm}(w)\,U^{la}(w')\,\rho^e(x')\,\rho^m(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{db}(z)\,U^{ae}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ld}(x')\,U^{me}(w')\,U^{am}(w)\,\rho^h(x')\,\rho^c(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{le}(x)\,U^{md}(w)\,U^{ma}(w')\,\rho^c(x')\,\rho^h(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{be}(x)\,U^{ad}(w)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{md}(x)\,U^{mb}(z)\,U^{ah}(w)\,\rho^l(x')\,\rho^e(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hd}(x)\,U^{lm}(w')\,U^{al}(w)\,\rho^m(x')\,\rho^e(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{le}(x)\,U^{md}(w)\,U^{lb}(z)\,U^{ma}(w')\,\rho^c(x')\,\rho^h(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{be}(x)\,U^{ad}(w)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{db}(z)\,U^{ae}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ld}(x')\,U^{me}(w')\,U^{am}(w)\,\rho^h(x')\,\rho^c(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{mb}(z)\,U^{ah}(w')\,\rho^{e}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(w)\,U^{hb}(z)\,U^{la}(w')\,\rho^{e}(x')\,\rho^{m}(x)\Big]
\end{aligned}
\tag{8.18}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'}e^{-ik(w'-w)}\Big[\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hl}(x)\,U^{md}(w')\,U^{am}(w)\,\rho^{l}(x)\,\rho^{e}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w')\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hl}(x)\,U^{md}(w)\,U^{ma}(w')\,\rho^{l}(x)\,\rho^{e}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(x)\,U^{ad}(w)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hl}(x)\,U^{md}(w')\,U^{am}(w)\,\rho^{l}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w')\,\rho^{m}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(w)\,U^{hb}(z)\,U^{ma}(w')\,\rho^{l}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bl}(x)\,U^{ad}(w)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\Big]
\end{aligned}
\tag{8.19}
$$

Row II We write it in two parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'}e^{-ik(w'-w)}\Big[\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{lm}(x')\,U^{bh}(z)\,U^{al}(w)\,\rho^{m}(x')\,\rho^{e}(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ca}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{eb}(z)\,U^{cd}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{bh}(z)\,U^{al}(w')\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ca}(w')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{db}(z)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{be}(z)\,U^{ad}(w)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ah}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{eb}(z)\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ae}(w')\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ah}(w')\,\rho^{c}(x')\,\rho^{e}(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x)\,U^{db}(z)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{lm}(x')\,U^{bh}(z)\,U^{la}(w)\,\rho^{m}(x')\,\rho^{e}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ca}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{eb}(z)\,U^{cd}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ea}(w')\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ah}(w')\,\rho^{c}(x')\,\rho^{e}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x)\,U^{db}(z)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{be}(z)\,U^{da}(w)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{eb}(z)\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{bh}(z)\,U^{la}(w')\,\rho^{m}(x)\,\rho^{e}(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ca}(w')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{db}(z)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{am}(w)\,\rho^{l}(x)\,\rho^{e}(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^{b}(x)\,\rho^{e}(x')\Big]
\end{aligned}
\tag{8.20}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Big[\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{bc}(x')\,\rho^{a}(x)\,\rho^{c}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{am}(w')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,\rho^{e}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{ma}(w)\,\rho^{l}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{bc}(x')\,\rho^{a}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(z)\,U^{bc}(x')\,\rho^{a}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{ma}(w')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,\rho^{e}(x')\,\rho^{b}(x)\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(z)\,U^{bc}(x')\,\rho^{c}(x')\,\rho^{a}(x)\Big]
\end{aligned}
\tag{8.21}
$$

Row III

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Big[\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(w')\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(x)\,U^{cl}(w')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(w)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ae}(x')\,U^{cl}(w)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(w')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')
\end{aligned}
$$

$$
\begin{aligned}
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{ad}(x')\,U^{ce}(w')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(w)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ae}(x')\,U^{cd}(w)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\Big]
\end{aligned}
\tag{8.22}
$$

Row IV Only 4 ρ's

Row V

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Big[\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(w')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{cd}(w)\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{cd}(w)\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{ce}(w')\,U^{bd}(x')\,\rho^{h}(x')\,\rho^{l}(x)\\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{ce}(w')\,U^{bd}(x')\,\rho^{h}(x')\,\rho^{l}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(w)\,U^{bd}(x')\,\rho^{h}(x')\,\rho^{l}(x)\\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(w')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{ce}(w')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(w)\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cd}(w)\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\Big]
\end{aligned}
\tag{8.23}
$$

Row VI

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{k^+}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Big[\\
\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{eb}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{bd}(z)\,U^{ce}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{cd}(x')\,\rho^{h}(x')\,\rho^{l}(x)
\end{aligned}
$$

$$
\begin{aligned}
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{eb}(z)\,\rho^h(x')\,\rho^l(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(z)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^h(x')\,\rho^l(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(x')\,\rho^h(x')\,\rho^l(x)\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Big]
\end{aligned}
\tag{8.24}
$$

8.4 The sum

Kernel structure by kernel structure, the colour structures of all terms above are reduced to an independent basis. The sum minus JIMWLK⊗LO is then:

kernel structure	class	independent colour structures	differences
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	charge–charge	12	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	charge–charge	10	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	charge–charge	13	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w,z)$	JIMWLK only	10	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w',w',z)$	JIMWLK only	10	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w',z)$	JIMWLK only	9	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	JIMWLK only	14	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	JIMWLK only	12	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	JIMWLK only	12	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	JIMWLK only	14	0
$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	charge–charge	14	0
$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	JIMWLK only	12	0
$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	JIMWLK only	12	0

Without the `4rho.tex` lists, 18 of 152 colour structures would differ (in $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$, $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$, $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$). The sum, collected, is JIMWLK⊗LO; in this form it reads

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{\text{JIMWLK}\otimes\text{LO}} = \mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^m(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^m(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{ad}(z)\,U^{bh}(x)\,U^{el}(x')\,\rho^l(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{ah}(x)\,U^{dl}(w')\,U^{eb}(z)\,\rho^l(x')\,\rho^c(x)\\
&+2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(x')\,U^{lh}(x)\,U^{da}(z)\,\rho^l(x)\,\rho^c(x')\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{lc}(x')\,U^{be}(x)\,\rho^h(x)\,\rho^l(x')\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w)\,U^{lm}(w')\,U^{lb}(w)\,\rho^m(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w)\,U^{lm}(w')\,U^{le}(w)\,\rho^m(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w)\,U^{lm}(x')\,U^{lb}(w)\,\rho^m(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w)\,U^{lm}(x')\,U^{le}(w)\,\rho^m(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w')\,U^{lb}(w')\,U^{lm}(w)\,\rho^c(x')\,\rho^m(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w')\,U^{le}(w')\,U^{lm}(w)\,\rho^c(x')\,\rho^m(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w')\,U^{lb}(w')\,U^{lm}(x)\,\rho^c(x')\,\rho^m(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w')\,U^{le}(w')\,U^{lm}(x)\,\rho^c(x')\,\rho^m(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{la}(w')\,U^{be}(z)\,U^{cd}(w)\,\rho^l(x')\,\rho^h(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{la}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ah}(w')\,\rho^b(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{be}(z)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x')\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{bd}(z)\,U^{cl}(w)\,\rho^l(x)\,\rho^h(x')\Big]
\end{aligned}
\tag{8.25}
$$

This is $\mathcal{N}\log\frac{k^+}{\Lambda}\times\mathcal{J}$, Eq. (10.1), before the charge average. Its terms (green / purple): 28 / 20.

9 The two-ρ terms $\propto\log\frac{\vee}{k^+}$: BFKL

The coefficient of $\log\frac{\vee}{k^+}$ of every row is its $\log\frac{\vee}{k^+}$ list of Secs. 3–5.

- It is zero for Group I Rows II (part 1) and V, and for $\mathcal{D}$.
- For the four-ρ terms, their three-ρ remainders and the `4rho.tex` lists it equals their $\log\frac{k^+}{\Lambda}$ terms.
- Group III Rows III, V, VI: the terms (B); Group II Row IV: H_4.

The two-ρ terms (genuine and from reordering the three- and four-ρ terms), row by row (orange / purple: 168 / 162):

Group I, Row II (three ρ;).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|_{I.II}^{\log\frac{\vee}{k^+}} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')
\end{aligned}
$$

$$
\begin{aligned}
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^{b}(x')\,\rho^{e}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^{c}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^{c}(x)\,\rho^{e}(x')\Big]
\end{aligned}
\tag{9.1}
$$

Group I, Row III (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{I.III} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\Big]
\end{aligned}
\tag{9.2}
$$

Group I, Row IV (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{I.IV} = \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x') \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x) \\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x) \\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x') \\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x) \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x) \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^m(x')\,\rho^e(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^m(x')\,\rho^e(x)\Bigg]
\end{aligned}
\tag{9.3}
$$

Group II, Row I (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{II.I} = \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)
\end{aligned}
$$

$$
\begin{aligned}
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^{c}(x)\,\rho^{e}(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^{c}(x)\,\rho^{b}(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x')\,\rho^{e}(x)\Bigg]
\end{aligned}
\tag{9.4}
$$

Group III, Row I (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\bigg|_{III.I}^{\log\frac{\vee}{k^+}} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{9.5}
$$

Group III, Row II (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{III.II} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^{b}(x')\,\rho^{e}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^{c}(x)\,\rho^{e}(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-2\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ha}(w')\,\rho^{c}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{9.6}
$$

Group III, Row III (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{III.III} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ca}(z)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{eh}(z)\,\rho^{d}(x)\,\rho^{c}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ca}(z)\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{eh}(z)\,\rho^{d}(x')\,\rho^{c}(x)\Big]
\end{aligned}
\tag{9.7}
$$

Group III, Row V (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{III.V} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')
\end{aligned}
$$

$$
\begin{aligned}
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\Big]
\end{aligned}
\tag{9.8}
$$

Group III, Row VI (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{III.VI} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ca}(z)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-2\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+2\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{eh}(z)\,\rho^{d}(x)\,\rho^{c}(x')\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ca}(z)\,\rho^{b}(x)\,\rho^{c}(x')\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dae}\,U^{hb}(x)\,U^{eh}(z)\,\rho^{d}(x')\,\rho^{c}(x)\Big]
\end{aligned}
\tag{9.9}
$$

Group II, Row I (genuine two ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{II.I^{\rm gen}} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x')\,\rho^{e}(x)\\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x')\,\rho^{c}(x)\,\rho^{l}(x')\\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{bc}(x')\,\rho^{a}(x)\,\rho^{c}(x')\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)
\end{aligned}
$$

$$
\begin{aligned}
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{ad}(w)\, U^{cl}(x')\, U^{be}(z)\, \rho^{l}(x')\, \rho^{h}(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{da}(w')\, U^{eb}(z)\, U^{hl}(x)\, \rho^{c}(x')\, \rho^{l}(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{bde}\, U^{ha}(w')\, U^{hd}(w)\, \rho^{c}(x')\, \rho^{e}(x)\Big]
\end{aligned}
\tag{9.10}
$$

Group II, Row IV (genuine two ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{II.IV^{\rm gen}} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+2\,N_c\;\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\, U^{bc}(x')\, \rho^{a}(x)\, \rho^{c}(x') \\
&+2\,N_c\;\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; \rho^{a}(x)\, \rho^{a}(x') \\
&-2\,N_c\;\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\, U^{cb}(x')\, \rho^{a}(x)\, \rho^{c}(x') \\
&-2\,N_c\;\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\, U^{cb}(x)\, \rho^{a}(x')\, \rho^{c}(x)\Big]
\end{aligned}
\tag{9.11}
$$

Group III, Row III, second part (genuine two ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{III.III^{\rm gen}} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{da}(w')\, U^{be}(z)\, U^{hl}(x')\, \rho^{c}(x)\, \rho^{l}(x') \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{da}(w')\, U^{be}(z)\, U^{hl}(x)\, \rho^{c}(x')\, \rho^{l}(x) \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{ad}(w)\, U^{eb}(z)\, U^{lc}(x)\, \rho^{l}(x)\, \rho^{h}(x') \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{ad}(w)\, U^{eb}(z)\, U^{lc}(x')\, \rho^{l}(x')\, \rho^{h}(x) \\
&-4\,N_c\;\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\, U^{bc}(x')\, \rho^{a}(x)\, \rho^{c}(x') \\
&+2\,N_c\;\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\, U^{cb}(x')\, \rho^{a}(x)\, \rho^{c}(x') \\
&+2\,N_c\;\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\, U^{cb}(x)\, \rho^{a}(x')\, \rho^{c}(x) \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dbe}\, U^{ha}(w)\, U^{eh}(z)\, \rho^{d}(x)\, \rho^{c}(x') \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dbe}\, U^{ha}(w)\, U^{eh}(z)\, \rho^{d}(x')\, \rho^{c}(x) \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dbe}\, U^{ha}(w')\, U^{eh}(z)\, \rho^{c}(x)\, \rho^{d}(x') \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dbe}\, U^{ha}(w')\, U^{eh}(z)\, \rho^{c}(x')\, \rho^{d}(x) \\
&+N_c\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\, U^{ac}(x')\, \rho^{c}(x')\, \rho^{b}(x) \\
&-N_c\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\, U^{ac}(x')\, \rho^{c}(x')\, \rho^{b}(x) \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{ae}(w')\, U^{cl}(x)\, U^{bd}(z)\, \rho^{l}(x)\, \rho^{h}(x') \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{ade}\, U^{hb}(w')\, U^{eh}(z)\, \rho^{d}(x)\, \rho^{c}(x') \\
&+N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\, U^{ac}(x)\, \rho^{b}(x')\, \rho^{c}(x) \\
&-N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x')\, U^{ac}(x)\, \rho^{b}(x')\, \rho^{c}(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{ad}(w)\, U^{cl}(x')\, U^{be}(z)\, \rho^{l}(x')\, \rho^{h}(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dae}\, U^{hb}(w)\, U^{eh}(z)\, \rho^{d}(x')\, \rho^{c}(x)\Big]
\end{aligned}
\tag{9.12}
$$

From four ρ's: remainders d^3N_a, d^3N_b.

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{4\rho,\;a\text{-}b} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\tfrac{1}{2}\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\, U^{ac}(w)\, \rho^{b}(x')\, \rho^{c}(x) \\
&-\tfrac{1}{2}\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\, U^{ac}(w)\, \rho^{b}(x')\, \rho^{c}(x) \\
&+\tfrac{1}{2}\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\, U^{ac}(x)\, \rho^{b}(x')\, \rho^{c}(x) \\
&-\tfrac{1}{2}\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\, U^{ac}(x)\, \rho^{b}(x')\, \rho^{c}(x) \\
&+\tfrac{1}{2}\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\, U^{ac}(w')\, \rho^{c}(x')\, \rho^{b}(x) \\
&-\tfrac{1}{2}\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\, U^{ac}(w')\, \rho^{c}(x')\, \rho^{b}(x) \\
&+\tfrac{1}{2}\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\, U^{ac}(x')\, \rho^{c}(x')\, \rho^{b}(x) \\
&-\tfrac{1}{2}\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\, U^{ac}(x')\, \rho^{c}(x')\, \rho^{b}(x) \\
&+\tfrac{1}{2}\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{cde}\, U^{hd}(x')\, U^{ha}(w)\, \rho^{b}(x)\, \rho^{e}(x') \\
&-\tfrac{1}{2}\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dce}\, U^{hd}(x)\, U^{ha}(w')\, \rho^{b}(x')\, \rho^{e}(x)
\end{aligned}
$$

$$
\begin{aligned}
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{1}{2}\;\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\;\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^{c}(x)\,\rho^{e}(x')\\
&-\tfrac{1}{2}\;\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\;\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^{c}(x)\,\rho^{e}(x')\\
&-N_c\;\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\;\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\tfrac{1}{2}\;\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\Big]
\end{aligned}
\tag{9.13}
$$

From four ρ's: the two-ρ lists of `4rho.tex`.

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{4\rho\text{-lists}} = \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-3\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+3\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^{m}(x')\,\rho^{e}(x)\\
&-3\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{ah}(w)\,U^{bl}(x)\,U^{dl}(z)\,U^{mh}(w')\,\rho^{m}(x')\,\rho^{e}(x)\\
&+3\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{dl}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^{m}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
&-2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^{b}(x')\,\rho^{e}(x)\\
&+\tfrac{1}{2}\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{3}{2}\;N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\tfrac{1}{2}\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^{e}(x')\,\rho^{c}(x)\Big]
\end{aligned}
\tag{9.14}
$$

kernel structure	class	independent colour structures	differences
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	charge–charge	12	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	charge–charge	10	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	charge–charge	13	0
$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	charge–charge	14	0
$\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$	BFKL only	14	0
$\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$	BFKL only	10	0
$\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$	BFKL only	13	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$	BFKL only	10	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$	BFKL only	13	0

The sum. Collected, the sum is BFKL⊗LO, Eq. (10.2) before the charge average (orange / purple: 23 / 12):

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|_{\text{BFKL}\otimes\text{LO}} = \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x)\,\rho^e(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,\rho^a(x)\,\rho^a(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x)\,\rho^d(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x')\,\rho^d(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ca}(z)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ca}(z)\,\rho^b(x')\,\rho^c(x)\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^b(x)\,\rho^c(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ca}(z)\,\rho^b(x)\,\rho^c(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ca}(z)\,\rho^b(x)\,\rho^c(x')\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ha}(w')\,\rho^c(x')\,\rho^e(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(x)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\Bigg]
\end{aligned}
\tag{9.15}
$$

10 The targets

In both formulas δY is the evolution interval: $\delta Y = \log\frac{k^+}{\Lambda}$ for JIMWLK and $\delta Y = \log\frac{\vee}{k^+}$ for BFKL. With $\langle\rho^a\rho^b\rangle = \delta^{ab}\langle\rho^f\rho^f\rangle/(N_c^2-1)$ and $\int_{w,w'} e^{-ik\cdot(w'-w)}$, $\mathcal{N}\,\delta Y \leftrightarrow \frac{\alpha_s^2\,\delta Y}{8\pi^6(N_c^2-1)}$.

10.1 JIMWLK⊗LO: your formula (App. F), corrected (M2)

With $P \equiv \frac{\alpha_s^2\delta Y}{8\pi^6(N_c^2-1)}$ and $\int \equiv \int_{x,y,t}\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\langle\rho^f(x)\rho^f(y)\rangle$, the rule is: in a term $\mathcal{K}(u,v,t)\,\mathrm{Tr}\{T^eU^\dagger(v)U(u)T^e\,M\}$,

$$M = (U^\dagger(u)-U^\dagger(t))(U(v)-U(t)),$$

i.e. M sits at the two points the colour charges act on. As sent, M was always $(U^\dagger(x)-U^\dagger(t))(U(y)-U(t))$ in $dN^{(1)}$, $dN^{(3)}$ and $(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))$ in $dN^{(2)}$, $dN^{(4)}$. The corrected formula:

$$\begin{aligned}
\frac{dN^{(1)}}{d^2w d^2w' dy} &= P\int\Big[\\
&\quad \mathcal{K}(y,x,t)\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))\}\\
&\quad -\mathcal{K}(w',x,t)\mathrm{Tr}\{T^eU^\dagger(x)U(w')T^e(U^\dagger(w')-U^\dagger(t))(U(x)-U(t))\}\\
&\quad -\mathcal{K}(y,w,t)\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(w)-U(t))\}\\
&\quad +\mathcal{K}(w',w,t)\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^e(U^\dagger(w')-U^\dagger(t))(U(w)-U(t))\}\Big],\\
\frac{dN^{(2)}}{d^2w d^2w' dy} &= P\int\Big[\\
&\quad \mathcal{K}(x,y,t)\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))\}\\
&\quad -\mathcal{K}(x,w',t)\mathrm{Tr}\{T^eU^\dagger(x)U(w')T^e(U^\dagger(w')-U^\dagger(t))(U(x)-U(t))\}\\
&\quad -\mathcal{K}(w,y,t)\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(w)-U(t))\}\\
&\quad +\mathcal{K}(w,w',t)\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^e(U^\dagger(w')-U^\dagger(t))(U(w)-U(t))\}\Big],\\
\frac{dN^{(3)}}{d^2w d^2w' dy} &= -P\int\Big[\mathcal{K}(y,y,t)\mathrm{Tr}\{T^e(U^\dagger(x)-U^\dagger(w))U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(y)-U(t))\}\\
&\quad -\mathcal{K}(w',w',t)\mathrm{Tr}\{T^e(U^\dagger(x)-U^\dagger(w))U(w')T^e(U^\dagger(w')-U^\dagger(t))(U(w')-U(t))\}\Big],\\
\frac{dN^{(4)}}{d^2w d^2w' dy} &= -P\int\Big[\mathcal{K}(x,x,t)\mathrm{Tr}\{T^eU^\dagger(x)(U(y)-U(w'))T^e(U^\dagger(x)-U^\dagger(t))(U(x)-U(t))\}\\
&\quad -\mathcal{K}(w,w,t)\mathrm{Tr}\{T^eU^\dagger(w)(U(y)-U(w'))T^e(U^\dagger(w)-U^\dagger(t))(U(w)-U(t))\}\Big],\\
\frac{dN^{(5)}}{d^2w d^2w' dy} &= -+P\int\Big[\\
&\quad \mathcal{K}(y,y,t)\mathrm{Tr}\{(U^\dagger(y)-U^\dagger(t))U(y)T^d\}\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))U(y)T^d\}\\
&\quad -\mathcal{K}(w',w',t)\mathrm{Tr}\{(U^\dagger(w')-U^\dagger(t))U(w')T^d\}\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))U(w')T^d\}\\
&\quad -\mathcal{K}(x,x,t)\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(t))U(x)T^d\}\mathrm{Tr}\{U^\dagger(x)(U(y)-U(w'))T^d\}\\
&\quad +\mathcal{K}(w,w,t)\mathrm{Tr}\{(U^\dagger(w)-U^\dagger(t))U(w)T^d\}\mathrm{Tr}\{U^\dagger(w)(U(y)-U(w'))T^d\}\Big],
\end{aligned}\tag{10.1}$$

and $\mathcal{J} = \sum_{i=1}^5 dN^{(i)}$ (per unit δY). After the correction, $dN^{(2)}$ is identical to $dN^{(1)}$ term by term. Numerically, at random points with random adjoint Wilson lines, $\sum_i dN^{(i)}$ equals JIMWLK⊗LO to 2×10^{-15}; as sent it differs by 0.22–0.47 (relative).

10.2 BFKL⊗LO: your formula (correct as sent)

$$\begin{aligned}
\left.\frac{dN}{d^2w\,d^2w'\,dy}\right|_{\mathrm{BFKL}} &= -\frac{\alpha_s^2N_c\,\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,z}\langle\rho^f(x)\rho^f(y)\rangle\Big[\\
&\quad \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')K_D(x,y,z)\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\\
&\quad -2\,\mathcal{K}(z-w)\cdot\mathcal{K}(z-w')K(x,y,z)\mathrm{Tr}\{(U^\dagger(z)-U^\dagger(w))(U(z)-U(w'))\}\\
&\quad +\mathcal{K}(z-w)\cdot\mathcal{K}(y-w')\big(2K(x,y,z)-K(x,x,z)\big)\mathrm{Tr}\{(U^\dagger(z)-U^\dagger(w))(U(y)-U(w'))\}\\
&\quad +\mathcal{K}(x-w)\cdot\mathcal{K}(z-w')\big(2K(x,y,z)-K(y,y,z)\big)\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(z)-U(w'))\}\Big],
\end{aligned}\tag{10.2}$$

with $K_D(x,y,z) = K(x,x,z) - 2K(x,y,z) + K(y,y,z)$. It agrees with BFKL⊗LO to 8×10^{-15}.

11 Checks

Scripts in `llsplit/`:

check	result
every evaluable term = its $\log \frac{k^+}{\Lambda}$, $\log \frac{\vee}{k^+}$ and finite parts, exactly in Λ, $\vee$ (`check_split.py`; 79 terms, 3 configurations, 2 sets of $k^+, \vee, \Lambda$)	1.5×10^{-14}
finite parts: exact remainder $\to$ its $\Lambda \to 0$ form	3.5×10^{-8} at $\Lambda = 10^{-9}$
$\frac{1}{\Lambda}$ lines of Group I Row III add up to zero; your two red lines	3.0×10^{-16}; 0
$\log \frac{\vee}{k^+} - \log \frac{k^+}{\Lambda}$ coefficient of each three-ρ row vs the exact row coefficient of the earlier check (`cross_check.py`)	$\leq 2.7 \times 10^{-13}$
poles $\times \log \frac{k^+}{\Lambda}$ cancel, with E1 / as written (`pole_check.py`)	8.2×10^{-16} / 0.5
three ρ's $\propto \log \frac{k^+}{\Lambda}$ / $\propto \log \frac{\vee}{k^+}$ cancel (`three_check.py`)	7.7×10^{-16} / 1.5×10^{-16}
the same with the edge kernel instead of $\mathcal{D}$	5.9×10^{-1}
two ρ's $\propto \log \frac{k^+}{\Lambda}$ vs JIMWLK$\otimes$LO; $\propto \log \frac{\vee}{k^+}$ vs BFKL$\otimes$LO, exact by colour structure (`../final_check/render.py`)	0 of 154; 0 of 109 differ
four ρ's cancel (`Final_evolution_check.pdf`)	4.7×10^{-15}